What is RAG (Retrieval-Augmented Generation)?

Definition

Retrieval-Augmented Generation (RAG) is an AI framework that enhances the capabilities of large language models (LLMs) by allowing them to access and incorporate information from external knowledge bases during the generation process. This helps LLMs to produce more accurate, up-to-date, and contextually relevant responses, mitigating issues like hallucination and outdated information.

How RAG Works

RAG typically involves two main phases: retrieval and generation. In the retrieval phase, when a user query is received, the system searches a vast external knowledge base (e.g., a database, a collection of documents, or the internet) for relevant information. This information is then passed to the LLM along with the original query. In the generation phase, the LLM uses both the query and the retrieved context to formulate a comprehensive and informed response.

Benefits

Improved Accuracy: Reduces the likelihood of generating incorrect or fabricated information.**Up-to-Date Information:** Allows LLMs to access the latest data beyond their training cutoff.**Reduced Hallucinations:** Grounds responses in factual, retrieved content.**Transparency:** Can often provide sources for the information used.**Domain Specificity:** Enables LLMs to perform well in specialized domains without extensive retraining.

Use Cases

Question Answering Systems: Providing precise answers to complex queries by drawing from extensive knowledge bases.**Content Creation:** Generating articles, reports, or summaries that are factually accurate and well-researched.**Customer Support Chatbots:** Delivering accurate and consistent information based on product manuals or FAQs.**Medical and Legal Research:** Assisting professionals in quickly finding and synthesizing relevant information from vast archives.